

Ministry of Consumer Affairs, Food & Public Distribution

1. Team Role & Responsibility Matrix (At a Glance)

2. Detailed Individual Contributions & Technology Dossiers

Domain: System Architecture

- **FastAPI Core Architecture (main.py):** Designed modular routing, asynchronous lifespan startup handlers, static asset mounting, and Jinja2 template rendering.
- **Real-Time WebSocket Engine (websocket_manager.py):** Developed connection manager supporting global broadcast channels, targeted token-subscriber queues, and center-specific telemetry streams.
- **5-Stage Procurement State Machine:** Implemented automated lifecycle tracking (REGISTERED -> GATE_ENTRY -> WEIGHBRIDGE -> QUALITY_CHECK -> PAYMENT_PROCESSED) ensuring tamper-proof state transitions.
- **Full-Stack Integration & Coordination:** Coordinated frontend-backend interface contracts, resolved cross-origin and real-time state synchronization across all modules.

Member #2: Backend Service & Queue Optimization Engineer

Domain: Dynamic Queue Wait-Time Algorithm

Executive Summary: Engineered mathematical wait-time heuristics, multi-mandi load balancing routing, and notification dispatch services.

Technologies: **Algorithmic Engine:** Python Heuristics, Mathematical Modeling (M/M/c queueing approximation); **Data Layer:** SQLite3 relational schemas, automated seed data generation, transactional logging; **Validation & Modeling:** Pydantic v2 (Strict Request Schemas & Dynamic Typing); **Notification Services:** Simulated Multi-Channel Messaging (SMS, WhatsApp, In-App Audit Trail); **Business Logic:** Official 2026 MSP rate indexing, Gross-to-Tare weight differential mathematics

Key Deliverables:

- **Dynamic Wait-Time Estimation Algorithm (queue_service.py):** Developed predictive calculation taking into account vehicle type factors (Tractors: 1.0x, Large Trucks: 1.8x, Pickups: 0.7x), crop lab durations, and active weighbridge lanes.
- **Mandi Load Balancer & Congestion Routing (centers.py):** Formulated intelligent routing recommending alternate nearby mandis when primary center wait time exceeds 45 minutes.
- **Database Architecture & Seeding (database.py):** Created optimized schema with tables for centers, farmers, procurement tokens, stage logs, and notification archives with 7+ realistic live demo records.
- **Multi-Channel Notification Dispatcher (notification_service.py):** Built automated SMS and stage milestone notification engine triggering alerts at every progression point.

Member #3: Frontend Lead & Farmer Experience Engineer

Domain: Farmer Slot Booking Portal

Executive Summary: Crafted the modern, farmer-centric UI/UX, responsive booking portal, dynamic progress tracker, and official e-Receipt vouchers.

Technologies: **Core Frontend:** HTML5, Modern JavaScript (ES6+ async/await, Fetch API, DOM APIs); **Styling & Design:** Tailwind CSS (Custom Kisan Emerald theme), CSS Grid, Responsive Flexbox; **Client State:** Local Storage persistence, dynamic DOM mutation, LiveSync WebSocket client; **Audio/Visual Feedback:** Web Audio API (Synthesized chime frequencies), Dynamic CSS Pulse Rings; **Print Engine:** Print CSS Media Queries for verified Govt e-Receipt & Payment Voucher generation

Key Deliverables:

- **Farmer Slot Booking Interface (farmer_book.html & farmer.js):** Built 3-step intuitive booking workflow with instant dynamic MSP payout calculator (Quantity x Rate).
- **Live 5-Stage Visual Tracker (farmer_track.html):** Created real-time progress bar, stage-specific active glow badges, vehicle-ahead counters, and dynamic wait-time indicators.
- **Official Verified e-Receipt / Voucher:** Designed printable and downloadable digital procurement voucher with official Govt headers, net quantity calculation, and Aadhaar DBT reference.
- **Accessibility '■■■■ ■■■■' (Simple Mode):** Implemented large-font high-contrast accessibility switch for elder farmers and low-vision users.

Member #4: Mandi Operations & Staff Workflow Engineer

Domain: Mandi Staff Operator Command Center

Executive Summary: Developed the operational dashboard, Kanban stage queue lanes, weighbridge gross/tare logging, and DBT approval triggers.

Technologies: **Dashboard Architecture:** JavaScript Event Dispatchers, Real-Time Kanban Pipeline, Dynamic Modals; **Mandi Operations UI:** Tailwind CSS Cards, Interactive Status Pills, Counter Badges; **Data Entry Modals:** Dynamic Context-Aware Forms (Gate, Scale, Moisture Lab, Tare Re-weigh); **Live State Polling:** Hybrid WebSocket sync with 1.5s background polling fallback; **Audio & Voice Triggers:** Automated bilingual voice approval announcements and stage chimes

Key Deliverables:

- **Mandi Operator Command Board (staff_dashboard.html & staff.js):** Built central operator interface with live counts for all 5 stages and mandi center switcher.
- **Kanban Pipeline Columns:** Developed drag/one-click transition lanes for Booked Slots, Gate Entry, Weighbridge, Quality Testing, and Completed DBT Payments.
- **Stage Data Capture Modals:** Built modal dialogs for digital scale weight inputs (Gross & Tare), moisture meter readings (FAQ Grade A standard <= 12%), and payment approvals.
- **DBT Payment Trigger:** Automated instant net weight computation (Net = (Gross - Tare)/100) and unique DBT transaction ID assignment upon final tare entry.

Member #5: Voice AI & Regional Accessibility Engineer

Domain: Multilingual i18n Engine

Executive Summary: Created the 5-language localization engine, Web Speech recognition voice assistant, spoken audio synthesis, and IVR dialer.

Technologies: **Localization Engine:** Custom Client i18n Engine supporting Hindi, Punjabi, Marathi, Telugu, English; **Speech Recognition:** Web Speech API (SpeechRecognition interface for Hindi / English input); **Speech Synthesis:** SpeechSynthesis API (Configured with native Indian accent, pitch & rate tuning); **Conversational Interface:** NLP Intent Extraction for token queries, crop MSP lookups, and queue wait times; **IVR Simulation:** DTMF Keypad simulation with audio tones, toll-free number simulation

Key Deliverables:

- **5-Language i18n Localization Engine (app.js):** Implemented instant client-side translation across the entire application with localStorage memory.
- **Interactive Voice Assistant '■■■■■ ■■■■■' (voice_assistant.js):** Built floating voice query assistant allowing farmers to speak queries naturally in Hindi or English.
- **Automated Audio Announcements:** Integrated automatic spoken status readouts (e.g. '■■■■■■! ■■■■ ■■■■ ■■■■■ ■■■■■■■■■ ■■■...') upon stage transitions.
- **Simulated Phone IVR System (ivr_demo.html & voice_ivr.py):** Developed basic-phone IVR simulator with keypad dialing for non-smartphone farmers.
- **Kisan AI Mandi Price Advisor (ai_advisor.html & ai_advisor.py):** Created AI-powered market advisory tool providing crop price trends and MSP comparisons.

Member #6: Ministry Analytics, QA & Live Simulation Engineer

Domain: Ministry Admin Analytics

Executive Summary: Built District & Ministry analytics dashboards, Chart.js throughput trends, live hackathon demo bots, and automated test suites.

Technologies: **Data Visualization:** Chart.js 4.x (Dynamic Line Charts, Doughnut Distribution, Bar Load Graphs); **Analytics Aggregation:** SQL Analytic Queries (SUM, AVG, Throughput per Hour, Metric Tons, Crores Disbursed); **Bottleneck Diagnostics:** SLA Threshold Monitoring (Optimal, Normal, Needs Attention warning badges); **Automated Demo Bot:** Auto Demo Script (auto_demo_bot.py) for continuous live presentation playback; **QA & Integration Tests:** End-to-End Automated Test Suites (test_pipeline.py, test_system.py)

Key Deliverables:

- **Ministry Admin Command Center (admin_dashboard.html & admin.js):** Built executive dashboard with live KPIs: Total Procured (Metric Tons), Total Disbursed (Crores), Active Vehicles in Queue.
- **Interactive Visualizations:** Developed real-time charts for Hourly Farmer Arrivals vs. Processed Output, Crop Distribution breakdowns, and Mandi Wait Time benchmarks.
- **Bottleneck Detection Table:** Created automated SLA tracking table diagnosing stage delays (e.g. Quality Lab inspection delays) with color-coded alerts.
- **Live Demo Simulation Engine (admin.py & auto_demo_bot.py):** Built 1-click simulation trigger for hackathon presentations to showcase live queue progression automatically.
- **Comprehensive Test Suites:** Developed automated end-to-end regression tests verifying all 5 stage transitions, weight deductions, and API responses.